

Ferrous Ammonium Sulfate

- Ferrous Ammonium Sulfate, 6-Hydrate,
 $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$
 Also: Ferrous Ammonium Sulfate Hexahydrate, (FAS)
 Reagent Grade, ACS Specifications

ECN-2-3275-2, ECN-0007/1

ECP-0007/1

- Sulfuric Acid, H_2SO_4
 Reagent Grade, ACS Specifications

Caution

Observe safety precautions for handling concentrated acids. Wear eye protection and impervious gloves. Use caution and always add acid slowly to water. Corrosive to skin, metals, and clothing. Avoid contact with liquid and vapor.

Solution Strengths

0.25 N	ECN-0024/1
	ECP-0026/1
	ECR-1125B
0.10 N	ECR-1113D
0.001 N	ECN-2-ML-3255

Preparation**0.25 N Ferrous Ammonium Sulfate**

- Add about 800 mL of distilled water to a 1-litre volumetric flask; stir on a magnetic stirrer.
- Add and dissolve 98.00 g of ferrous ammonium sulfate, 6-hydrate.
- Carefully add 20 mL of concentrated sulfuric acid.
- Cool flask under running water.
- Dilute to volume with distilled water; mix thoroughly.

0.10 N Ferrous Ammonium Sulfate

- Add about 300 mL of distilled water to a 1-litre volumetric flask; stir on a magnetic stirrer.
- Add and dissolve 39.20 g of ferrous ammonium sulfate, 6-hydrate.
- Carefully add 25 mL of 7.0 N sulfuric acid.
- Dilute to volume with distilled water; mix thoroughly.

0.001 N Ferrous Ammonium Sulfate

- Place a magnetic stir bar and about 300 mL of distilled water to a 500 mL volumetric flask. Place on a magnetic stirrer.
- Measure out 100 mL of 2.5 N sulfuric acid in a 100 mL graduated cylinder and add to the flask, while stirring.

3. Weigh 0.2 g of ferrous ammonium sulfate and transfer to the volumetric flask.
4. Dilute to volume with distilled water; mix thoroughly.

Note: This reagent is stable for four weeks.

Assay of FAS**Procedure Using Potassium Dichromate**

See Reagent Preparation for:

- Potassium Dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$, **NIST Oxidimetric Primary Standard**, SRM-136e (or subsequent lot of SRM-136)
- Ferroin Indicator
- Sulfuric Acid, concentrated
- Ferrous Ammonium Sulfate, 6-Hydrate (FAS)
 1. Weigh 14 g of FAS to the nearest 0.1 mg and record.
 2. Quantitatively transfer to a 200 mL volumetric flask with distilled water and swirl the flask to dissolve the FAS.
 3. Dilute the solution to the mark with distilled water. Invert the flask, 6 to 10 times, to mix thoroughly.
 4. Dry the potassium dichromate (SRM-136e) for 2 hours at 110°C and cool to room temperature. Weigh 1 g to the nearest 0.1 mg and record.
 5. Quantitatively transfer to a 100 mL volumetric flask containing 50 mL of distilled water. Swirl the flask to dissolve the potassium dichromate.
 6. Dilute the solution to the mark with distilled water. Invert the flask, 6 to 10 times, to mix thoroughly.
 7. Fill a 50 mL buret with the FAS solution (Step 3, above).
 8. Add 20 mL concentrated sulfuric acid to a 150 mL beaker containing 25 mL of distilled water and a magnetic stirring bar.

Caution

Hot acid solution is formed.

9. Pipet 15.0 mL of the potassium dichromate solution into the beaker.
10. Add 2 drops of ferroin indicator to the beaker and place the beaker on a magnetic stirrer.
11. Slowly add the FAS solution from the buret to the beaker. The solution will change colors as the titrant is added, going from a *green-blue* color to a *reddish-brown* color. The point that produced the first *reddish-brown* color change is the end point.
12. Record the volume of FAS titrant consumed to the nearest 0.05 mL at the end point.
13. Repeat Steps 7 to 12, two more times.

Calculations:**N of Potassium Dichromate**

$$N \text{ K}_2\text{Cr}_2\text{O}_7 = \frac{(g, \text{K}_2\text{Cr}_2\text{O}_7)(1000)}{(100 \text{ mL})(294.2/6)} = 0.2039(g, \text{K}_2\text{Cr}_2\text{O}_7)$$

Where:

- g, $\text{K}_2\text{Cr}_2\text{O}_7$ = recorded weight from Step 4
 1000 = factor to convert eq wt to meq wt
 100 mL = volume of $\text{K}_2\text{Cr}_2\text{O}_7$ solution
 294.2 = the molecular wt of $\text{K}_2\text{Cr}_2\text{O}_7$
 6 = factor to convert g-mole wt to eq wt of $\text{K}_2\text{Cr}_2\text{O}_7$

N of Ferrous Ammonium Sulfate, 6-Hydrate (FAS)

$$N \text{ FAS} = \frac{(N \text{ K}_2\text{Cr}_2\text{O}_7)(\text{mL K}_2\text{Cr}_2\text{O}_7)}{(\text{mL FAS})} = \frac{15.0(N \text{ K}_2\text{Cr}_2\text{O}_7)}{(\text{mL FAS})}$$

Where:

- N $\text{K}_2\text{Cr}_2\text{O}_7$ = normality (meq/mL) calculated, above
 mL $\text{K}_2\text{Cr}_2\text{O}_7$ = 15.0 mL volume of $\text{K}_2\text{Cr}_2\text{O}_7$ solution taken
 mL FAS = average* volume of the three titrations required to reach the end point color change.

* The average volume for the three titrations should not have a standard deviation (1s) greater than 0.050 mL.

g/L, Iron(II) in Ferrous Ammonium Sulfate

$$g/L, \text{Iron(II)}_{\text{actual}} = 55.85 (N \text{ FAS})$$

Where:

$$55.85 = \text{eq wt (g-eq) of Iron(II) in FAS}$$

Theoretical wt of Iron(II) in Solution

$$g/L, \text{Iron(II)}_{\text{theoretical}} = \frac{(g, \text{FAS})(55.85)(1000)}{(200 \text{ mL})(391.85)} = 0.7139(g, \text{FAS})$$

Where:

- g, FAS = g-weight $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$ recorded in Step 1
 55.85 = atomic weight of iron (g-mole)
 1000 mL = factor to convert mL to L
 200 mL = dilution vol of wt of $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$
 391.85 = mole wt (g-mole) of ferrous ammonium sulfate, 6-hydrate.

Assay Percentage

$$\% \text{ Iron(II)} = \frac{[g/L, \text{Iron(II)}_{\text{actual}}](100)}{[g/L, \text{Iron(II)}_{\text{theoretical}}]}$$

ECN-0007/1**ECN-2-3275-2****ECP-0007/1****Alternate Procedure Using Potassium Permanganate**

See Reagent Preparation for:

- *Potassium Permanganate*, KMnO_4 ,
- *Reagent Water*
- *Sulfuric Acid*, concentrated
- Ferrous Ammonium Sulfate, 6-Hydrate (FAS)

Note: This procedure is based on the method in Reagent Chemicals, 8th Edition, American Chemical Society, 1993.

Note: This procedure should be repeated in triplicate with the average of the three results used as the assay value.

1. Accurately weigh 1.6 grams of sample to the nearest milligram and dissolve in a mixture of 100 mL reagent water and 3 mL of concentrated sulfuric acid contained in a 250 mL Erlenmeyer flask.

**Caution**

Always add acid to water and not water to acid.

2. Titrate while stirring with standardized 0.1 N potassium permanganate (KMnO_4) from a 50 mL buret to a permanent faint pink endpoint that lasts for at least 15 seconds.
3. Repeat steps 1 and 2 without any sample. This is the blank.

Calculations

$$\% (\text{wt./wt.}) \text{ ferrous ammonium sulfate, hexahydrate} = \frac{(\text{mL KMnO}_4 \text{ sample} - \text{mL KMnO}_4 \text{ blank}) \times N \text{ KMnO}_4 \times 0.3921 \times 100}{\text{sample size in grams}}$$

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